JIEUN SUNG

Ph.D. Candidate in Bioinformatics | AI for Drug Discovery & Computational Biology

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EDUCATION

Ewha Womans University **Sep 2021-Feb 2027 (Expected)**
Ph.D. Candidate in Life Science, Specialization in Bioinformatics
- Systems Pharmacology Laboratory, Department of Life Science

Ewha Womans University **Mar 2017-Aug 2021**
B.S. in Chemistry with a Double Major in Bioinformatics
Honors: Graduated with Honors (GPA 3.84/4.3); Academic Excellence Scholarship

RESEARCH INTERESTS

Biological foundation models; single-cell and perturbation modeling; transcriptomics and gene regulation; computational target discovery; AI for drug discovery.

SELECTED AI RESEARCH HIGHLIGHTS

***scDEBART* - Single-Cell Perturbation Prediction.** Most single-cell foundation models learn static cell states rather than how cells respond to perturbations. scDEBART is pretrained on 6.28M differential-expression profiles to learn how gene expression changes across cellular states and achieves 4–7× higher recovery of perturbation-responsive genes than existing models. **Published at ICML 2026.**

***TDEP* - Interpretable Drug–Target Interaction Prediction.** Drug-induced transcriptional responses reflect the combined effects of multiple protein targets. TDEP uses known drug–target interactions to disentangle these mixed responses into target-specific gene-expression effects, enabling prediction of new drug–target interactions with **gene- and pathway-level interpretation of their cellular effects**, which conventional DTI models do not provide.

***FateDriver* - Weakly Supervised Dual-View Graph Learning for Transition-Specific Prioritization of Candidate Regulators in Cellular Reprogramming.** Cellular reprogramming converts one cell state into another, but the endogenous genes that actively drive this transition remain poorly characterized. FateDriver jointly models cell-state transitions and gene-regulatory relationships with a dual-view graph neural network to **identify candidate regulators that promote the desired cell fate.**

PUBLICATIONS

1. **Sung, J.** and Kim, W. (2026). scDEBART: Predicting *in silico* Single-Cell Perturbation Responses via Large-Scale Differential Expression Learning. In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*. OpenReview.
2. **Sung, J.**, Kim, S., and Kim, W. (2026). TDEP: Predicting Target Perturbation Response from Drug-Induced Transcriptomes. Preprint, Research Square (under review at BMC Bioinformatics). https://doi.org/10.21203/rs.3.rs-10699182/v1
3. **Sung, J.***, Son, K.*, Cha, H.-J., Kim, W., and Lee, H. (2026). FateDriver: Weakly supervised dual-view graph learning for transition-specific prioritization of candidate regulators in cellular reprogramming. Preprint, bioRxiv. https://doi.org/10.64898/2026.10.27.762150
4. **Sung, J.** and Kim, W. TDEM: Deconvolving Drug-Induced Transcriptomes into Target-Specific Perturbation Profiles via Matrix Factorization. Manuscript in preparation.

* Equal contribution.

PATENTS

1. Kim, W. and **Sung, J.**, Method for Constructing a Model for Predicting Transcriptomic Expression Changes, Method for Predicting Transcriptomic Expression Changes Using a Trained Model, and Hardware Device. Republic of Korea patent application No. 10-2026-0167782, filed September 3, 2026. Assignee: Ewha Womans University.
2. Kim, W. and **Sung, J.**, Method for Predicting Candidate Genes Inducing Transcriptomic Expression Changes, Method for Predicting Candidate Gene Sets Inducing Transcriptomic Expression Changes, and Hardware Device. Republic of Korea patent application No. 10-2026-0167802, filed September 3, 2026. Assignee: Ewha Womans University.

HONORS & AWARDS

Jul-Aug 2025 **Third Place, 3rd AI Drug Discovery Competition**
Placed 3rd among 1,137 participants in a national competition focused on predicting MAP3K5 inhibitor IC50 values from molecular structures. Hosted by the Korea Pharmaceutical and Bio-Pharma Association and sponsored by the Ministry of Health and Welfare, Yuhan Corporation, and CAS.

Aug-Sep 2024 **Second Place, 2nd AI Drug Discovery Competition**
Placed 2nd among 1,798 participants in a national competition focused on predicting IRAK4 inhibitor IC50 values from molecular structures. Hosted by the Korea Pharmaceutical and Bio-Pharma Association's AI Drug Convergence Research Institute and sponsored by the Ministry of Health and Welfare, Korea Health Industry Development Institute, and Daewoong Pharmaceutical.

RESEARCH FUNDING

National Research Foundation of Korea (NRF) Ph.D. Student Research Promotion Grant

AI-Based Modeling of Drug-Induced Transcriptomic Changes

Funding period: Sep 2024–Aug 2026

Principal Investigator | Total funding: KRW 50,000,000.

TECHNICAL EXPERTISE

Programming & ML Frameworks: Python, R, Bash, Git, SLURM; PyTorch, TensorFlow, Keras, Hugging Face Transformers, DeepSpeed
Machine Learning: Transformer architectures, graph neural networks, autoencoders, representation learning, multi-GPU distributed training, large-scale model pretraining
Computational Biology: Single-cell RNA-seq analysis (Scanpy, AnnData, scvi-tools), Bulk RNA-seq analysis (FastQC, fastp, STAR, Salmon, featureCounts, HTSeq), perturbation transcriptomics (Perturb-seq; PerturbScore), differential expression analysis(DESeq2, edgeR, limma), network analysis (STRING, NicheNet), pathway enrichment(clusterProfiler, fgsea), Visualization (ComplexHeatmap, ggplot2)
Computational Drug Discovery: Molecular property prediction, drug–target interaction prediction, molecular docking, virtual screening, protein structure prediction (AlphaFold)

PRESENTATIONS

Invited Talks

1. **Sung, J.** Building AI Prediction Models from Biological Data. Invited lecture, Systems Biology Lab, Department of Pharmacy, Pusan National University, Busan, Republic of Korea, February 2026.
2. **Sung, J.** AI-Based Virtual Cell Modeling for Drug Discovery. Invited lecture, undergraduate course, Department of AI-Based Drug Development, Duksung Women's University, Seoul, Republic of Korea, September 2026.

Contributed Talks and Posters

1. **Sung, J.**, Son, K., Cha, H.-J., Kim, W., and Lee, H. FateDriver: A Dual-View Graph Neural Network for Predicting Driver Genes in TF-Induced Cellular Reprogramming. Annual Meeting of the Korean Society for Stem Cell Research (KSSCR), Republic of Korea, August 2026. Poster.
2. **Sung, J.** and Kim, W. scDEBART: Predicting *in silico* Single-Cell Perturbation Responses via Large-Scale Differential Expression Learning. 43rd International Conference on Machine Learning (ICML 2026), COEX Convention & Exhibition Center, Seoul, Republic of Korea, July 2026. Poster.
3. **Sung, J.**, Kim, S., and Kim, W. TDEP: Target-Perturbed Differential Expression Profiles for Interpretable Drug–Target Interaction Prediction. Annual Meeting of the Korean Society for Bioinformatics (KSBI), Republic of Korea, October 2025. Poster.
4. **Sung, J.** and Kim, W. Uncertainty-Aware BLAST-Weighted Multi-Representation Ensemble Model for IC50 Prediction. 3rd AI Drug Discovery Competition, Republic of Korea, September 2025. Poster.

5. **Sung, J.**, Kim, S., and Kim, W. AI-Based Deconvolution of Drug-Induced Transcriptomic Changes into Single-Target Effects across Multiple Cell Lines. EJO Symposium, Republic of Korea, December 2024. Poster and oral presentation.
6. Six additional posters at domestic conferences (Korea Genome Organization; Korean Society for Bioinformatics; Korean Society for Biochemistry and Molecular Biology; Korean Society for Aging Research; 2nd AI Drug Discovery Competition), Republic of Korea, 2022–2026.